

Research Introduction

M1 Kazuya Okamura

2024/07/25

Introduction

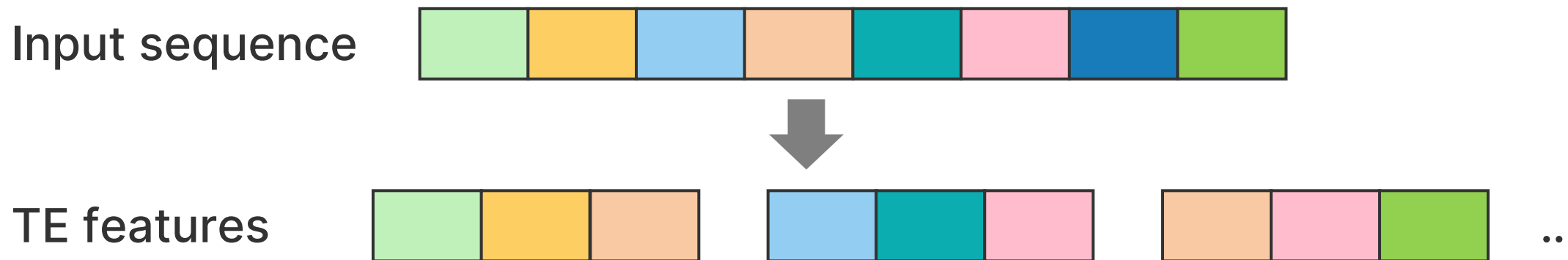
- Joint research with JRI (日本総研) researcher
 - Same as M1 Suto
- Research Theme:
 - Deep Randomized Time Warping for Action Recognition ^[1]
 - Mechanism of Randomized Time Warping versus Attention Module ^[2] (not yet published)

↓
- Application based on RTW method

Overview of methods

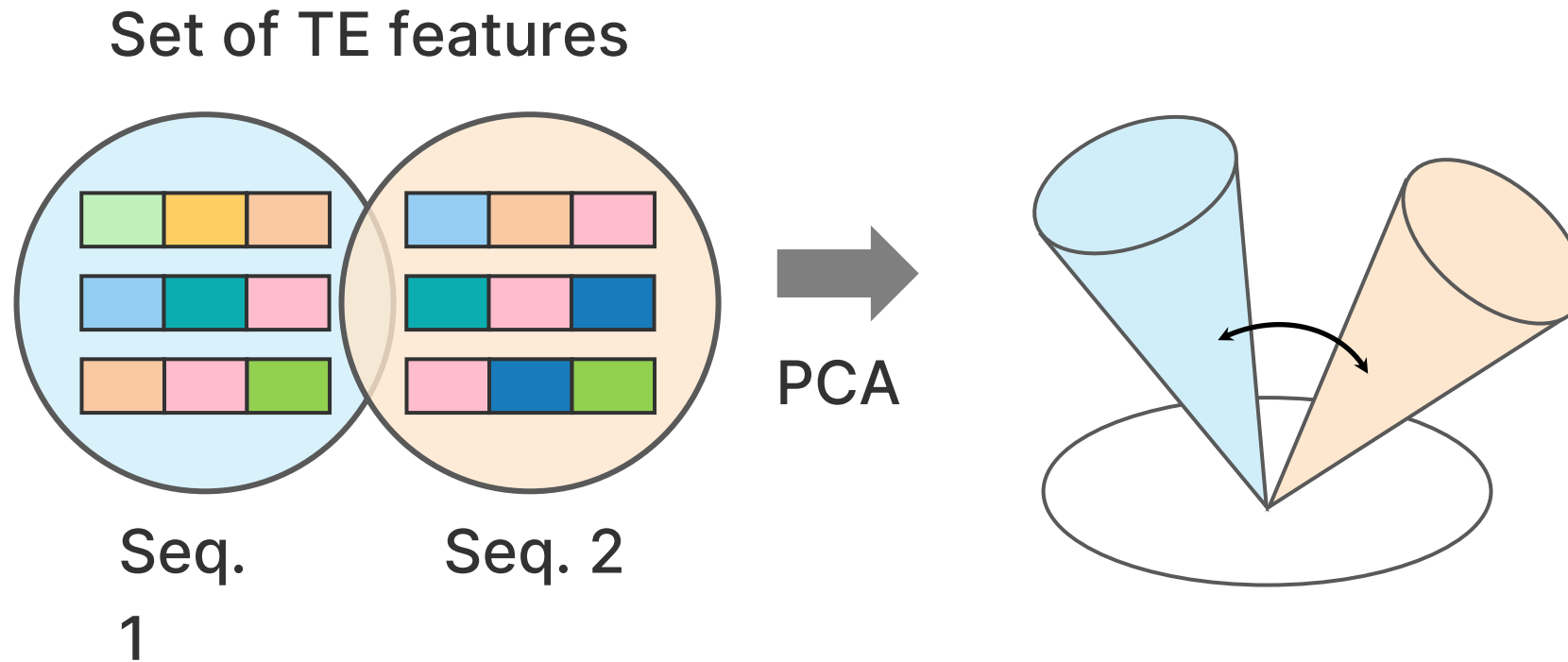
Randomized Time Warping (RTW)

- An extension method of DTW (Dynamic Time Warping)
- **Time elastic (TE) features:**
Random sampling of input sequences while preserving temporal features



■ Randomized Time Warping (RTW)

- Applying PCA to a set of TE features, measured by canonical angles, is more effective than DTW



Randomized Time Warping (RTW)

Advantage

- By random sampling, TE features can be represented at **local** and **global scales**
- RTW can be interpreted as focusing on highly similar features among multiple TE features
 - It can be regarded as a **self-attention mechanism**
- Furthermore, subspace representations can be generated stably even with small amounts of data
 - In small sample sizes, it is superior to self-attention mechanism

Overview of methods

- Deep Randomized Time Warping for Action Recognition ^[1]
 - Extracted from 2D/3D CNN and converted to TE features through RTW
 - Therefore, it can be applied to various CNN architectures
- Mechanism of Randomized Time Warping versus Attention Module ^[2]
 - Spatial Transformer + RTW
 - RTW performs as well as or better than a self-attention mechanism

Future work

- Reproduce the Transformer+RTW program and verify that it works properly on supercomputer
- Consider applications based on RTW method (not just video recognition)
 - To other targets
 - Sign language recognition
 - etc...

Progress

- Verify Transformer+RTW program on supercomputer

Done

- ✓ Verification of GPU training behavior using previous research programs
- ✓ Dataset preparation (Something-Something V2)

Verification of GPU training behavior

- ✓Whether GPU-based learning is possible
- ✓Whether the learning is progressing correctly
 - Using PSPNet (Pyramid Scene Parsing Network)

```
bnode120                      Wed Jul 24 18:56:53 2024  550.90.07
[0] NVIDIA H100 PCIe | 65°C, 100 % | 11987 / 81559 MB | kokamura(11978M)
```

```
Totally 12555 samples in train set.
Totally 80 samples in test set.
Epoch [1/30]
Train: 100%|#####| 784/784 [05:00<00:00, 2.61it/s, mIoU=0.943, mAcc=0.968]

loss      mIoU      mAcc      allAcc
=====
0.059     0.837     0.906     0.977

Testing...
Test: 100%|#####| 80/80 [00:05<00:00, 14.85it/s]

mIoU      mAcc      allAcc
=====
0.836     0.904     0.977
```

Verification of GPU training behavior

- PSPNet

Training speed

GPU / Computer	Training speed
NVIDIA RTX2060 (6GB) on laptop computer	30m /epoch
NVIDIA H100 (80GB) on supercomputer	4-5m /epoch

Dataset

- 12,555 images in total

Hyperparameter

- Batch size : 16
- Input image : 473x473

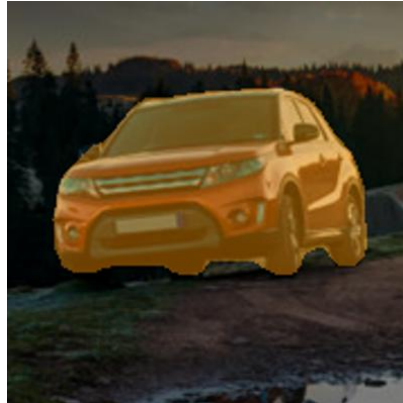
■ Verification of GPU training behavior

- PSPNet
 - Results of 5 epoch

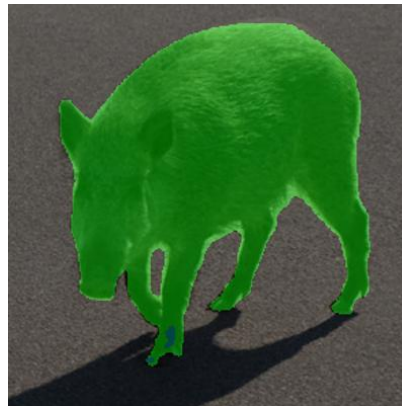
airplane



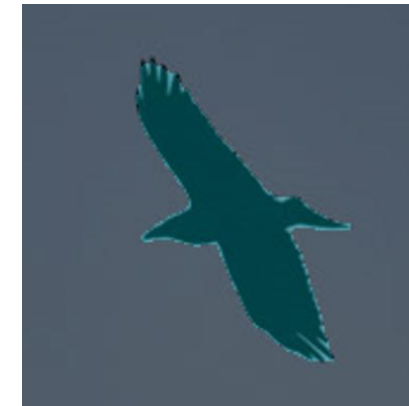
car



boar



crow

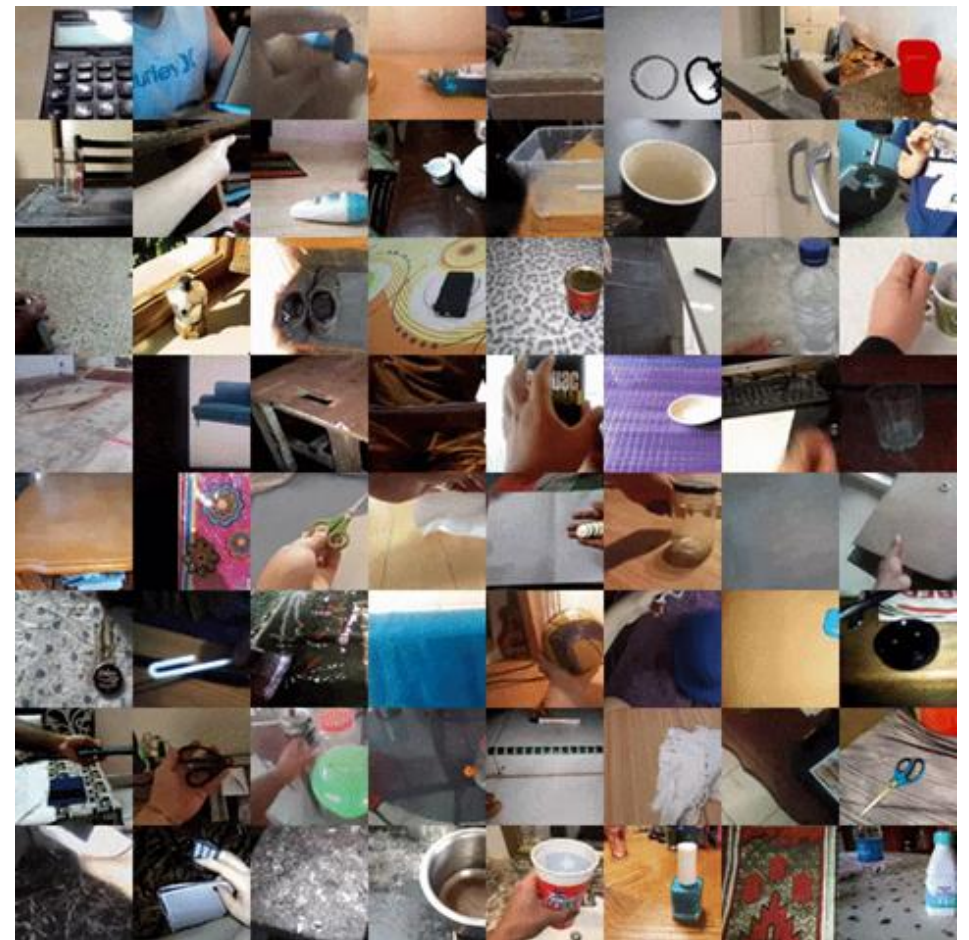


Dataset preparation

- Something-Something V2
 - Moving objects dataset

Dataset details

Total number of videos	220,847
Training Set	168,913
Validation Set	24,777
Test Set (w/o labels)	27,157
Labels	174
Quality	100px
FPS	12



- ✓ Extract frames from videos
- ✓ Resize images

Progress

- Verify Transformer+RTW program on supercomputer

Done

- ✓ Verification of GPU training behavior with the PyTorch program (using previous research programs)
- ✓ Dataset preparation (Something-Something V2)

Next

- Reproduce Transformer+RTW program and experiment to obtain similar results.
- Consider application methods
- **personal** Revise paper on research at previous school

Supercomputer Specs (Pegasus)

CPU	Intel(R) Xeon(R) Platinum 8468 (2.1GHz/48 コア) ×1
メモリ	DDR5-4800 16GiB DIMM ×8 ※動作は 4400MHz DDR5-4800 256GiB Persistent Memory ×8
ストレージ	NVMe 3.2TB 3DWPD ×2
インタフェース	InfiniBand NDR200 HCA ×1
GPU	NVIDIA H100 ×1

▼ NVIDIA H100

Tensor 倍精度性能	51 TFLOPS
メモリ	80GiB
メモリバンド幅	2TB/sec
バスインタフェース	PCIe Gen5